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Lighbrary



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I. Abstract

Libraries face lighting-related issues, including glare caused by shifting sunlight angles, energy waste from continuous lighting, and uneven light intensity that strains eyes. To address these challenges, this project aims to develop an IoT-Based Library Adaptive Lighting Environment System. The core procedure involves connecting sensors for real-time light detection, assembling and coding an Arduino AVR for automatic adjustment, and program a touchable screen . Technically, the system will auto-detect glare to block harsh sunlight via adjustable shades, dynamically dim or brighten smart bulbs based on real-time light conditions, and eliminate uneven lighting distribution—resolving the aforementioned problems. This integrated solution not only enhances user comfort and work but also delivers significant social and environmental impacts: it reduces unnecessary energy consumption to promote sustainability, lowers operational costs, and mitigates eye fatigue and disease risks for library users, contributing to the development of eco-friendly and comfortable library spaces.



II. Acknowledgements

Thanks to the TAs (Yu Lu, Rui Li, Yue Wang, Zhenyou Huang, Chiheng Zhu), professors (Prof. Pi, Prof. Wei), our teammates (Congyu Yang, Chen Liang, Shuhan Guo, Kuangdi Li), and others who helped with our project.

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III. Problem

Libraries, as essential public spaces, rely heavily on adequate lighting to ensure a comfortable environment for reading and study. However, most libraries still utilize traditional lighting control methods, such as manual switches, which fail to dynamically respond to changing natural light conditions. This outdated approach leads to several problems:

1) Significant Energy Waste: Artificial lights are often kept on at full capacity throughout operating hours, even when sufficient daylight is available. This results in unnecessary electricity consumption with higher operational costs and a larger carbon footprint. This also leads to increased maintenance efforts and missed opportunities for energy savings.

2) Suboptimal Reading Experience: Fixed lighting levels cannot adapt to varying natural light throughout the day, leading to problems such as glare on screens or insufficient illumination on reading materials, as shown in Figure 1. This causes visual discomfort and fatigue for eyes, negatively impacting their concentration and ability to study for extended time.



Figure 1: glare

3) Lack of Automated Responsiveness: Current systems depend on users to manually adjust lights, which is inefficient and often inconsistent. There is no integrated mechanism to automatically block natural light and complement it with artificial lighting in real-time.

Studies indicate that lighting constitutes a significant portion of energy consumption in public and commercial buildings. Real-world implementations have demonstrated that intelligent systems which block daylight and incorporate occupancy sensors can achieve substantial energy savings; the Hayden Library project, for example, reduced lighting electricity consumption by approximately 43 percent. Further supporting this, research on smart dimmable LED systems that optimize daylight use and respond to occupancy has shown the potential to reduce energy consumption by about 40%. Broader industry analyses also indicate that advanced LED lighting systems with integrated controls can achieve even higher energy savings, potentially up to 75%, by automatically adjusting light levels based on available daylight and occupancy.

While smart lighting solutions exist, they are often designed for offices or homes and lack the specific features needed for libraries, such as uniform light distribution and glare control. Their high cost and complexity also limit adoption in public institutions. Our project addresses this gap by developing a more targeted and cost-effective system for library environments.

In summary, the current library lighting infrastructure is characterized by:

1. Inefficient energy usage due to bad integration and utilization of available natural light effectively.
2. Poor user experience without consistent, comfortable, and adaptive illumination.
3. Low level of automation that requires manual operations for basic lighting adjustments.
4. High cost and complexity of existing smart solutions, limiting their feasibility for widespread library implementation.

IV. Needs

To solve the lighting problems in current libraries, the following materials, methods, and tasks are required:

To address sunlight glare, the system can auto-detect glare, block harsh sunlight, and cleverly utilize natural sunlight while adjusting indoor lighting in tandem.

2. For energy conservation purposes, it automatically dims or brightens indoor lights and runs only when required to cut down on unnecessary energy consumption.

3. By integrating the above two functions, the system is further optimized to seamlessly adapt to library scenarios, balancing user comfort and energy efficiency.

V. Solution ... Lighbrary

To address the lighting needs of library scenarios—balancing energy efficiency, optimal lighting comfort, and intelligent adaptability—we propose the Lighbrary intelligent lighting system, which entails:

1. Smart on/off with human detection: The system automatically turns on when someone enters the area and goes into energy-saving sleep mode when no one is around, avoiding waste of electricity.
2. Sensors to decide outdoor glare/indoor brightness: The system uses UV sensors outside to decide glaring extent and photoresistors to decide indoor lighting. These data are collected and sent to the MCU to proceed further calculations.
3. Automatic adjustment of blinds and lights: The system's MCU automatically controls blinds to block glare and adjusts LED lights' brightness to keep lighting comfortable.
4. Validation of the whole system: Deliberately designed experiments carried out, trying to provide users with better experiences.

The Lighbrary system is an integrated intelligent lighting solution for library environments, designed to autonomously adjust lighting conditions and reduce energy waste without manual intervention. Also, it allows users to manually adjust lighting and blinds if they need.

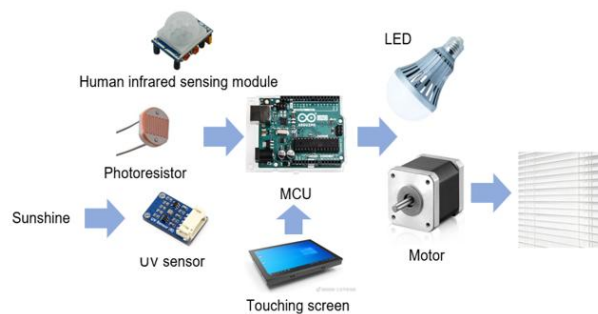


Figure 2 Lighbrary System Concept Diagram

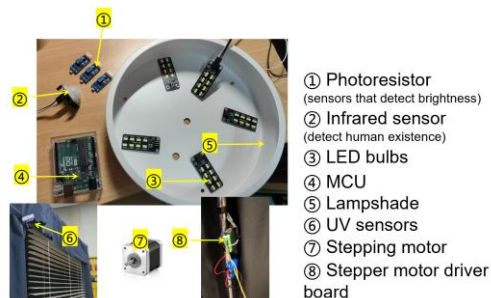


Figure 3 Key Components of the System

As shown in Figures 2-3, The Lighbrary system consists of the following core functional components: the human infrared sensing module, UV sensors, photoresistors, MCU control unit, mechanical execution unit (stepping motors connected with the blinds), a touching screen (not shown in Figure 3) and LED lightings. Each component is chosen and designed to address the key pain points of traditional library lighting systems.

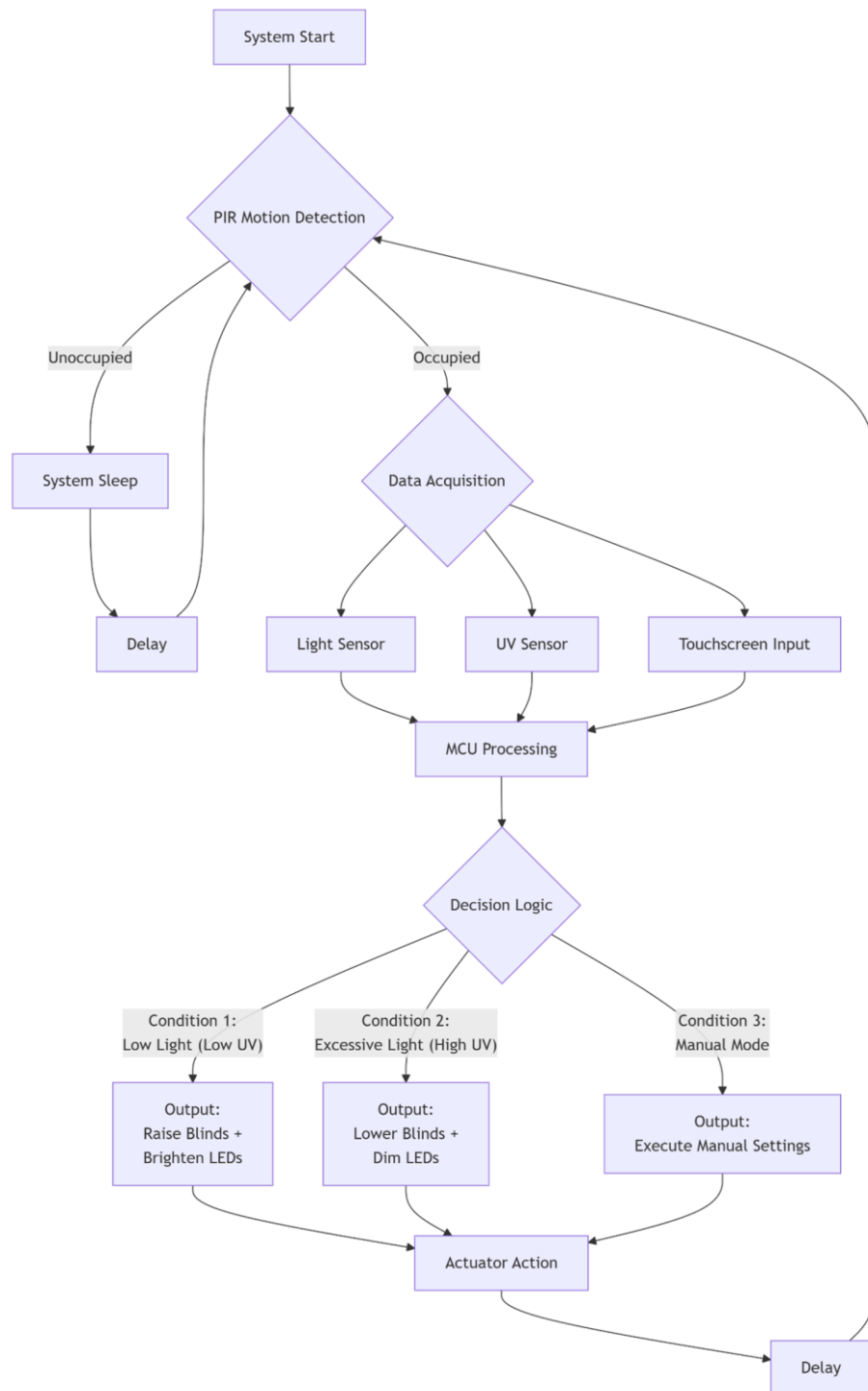


Figure 4 The Flow Diagram of the System's Control Logic

As illustrated in Figure 4, the whole system functions as follows:

First, the infrared sensor unit (main switch) is the key point of energy efficiency. It is placed at the entrance of the simulated library zone. When no human activity is detected within the sensing range (adjustable to 0-7 meters), the entire system enters sleep mode, with all non-essential components powered down, reducing standby power consumption. Once users enter the zone, the sensor immediately sends a wake-up signal to activate the full system, ensuring lighting is only available when needed.

Second, the UV sensor, together with photoresistors, improves the issue of unbalanced lighting caused by outdoor glare and uneven indoor lighting distribution. The outdoor UV sensor is installed next to the blinds on the walls, calibrated to detect the intensity of ultraviolet radiation—a direct indicator of sunlight glare. When UV intensity exceeds a preset threshold (adjustable based on library location and orientation), the sensor transmits a glare signal to the MCU. Meanwhile, three photoresistors are deployed at different key positions in the simulated library. By calculating the average value of the three photoresistors' readings, the system reduces detection errors caused by local light spots or shadows, ensuring accurate assessment of indoor lighting conditions.

The MCU control unit ensures intelligent and responsive operation. We use a MCU preloaded with deliberately designed control program shown in Figure 4. When the UV sensor detects strong glare over a certain number, the MCU outputs an analog signal to the motor to close the blinds incrementally (the closing degree is proportional to glare intensity); simultaneously, if the average light intensity from the photoresistors is below the standard reading desk illuminance, the MCU adjusts the LED lighting to the corresponding brightness level. The control logic supports on-site parameter modification through the touching screen, allowing users to tailor settings to specific needs. See detailed code in Appendix.

Finally, the output signals are transmitted to the LEDs and a stepping motor connected to the blinds. By honestly following the commands from the MCU, the whole system now is able to realize its basic functions. It's worth mentioning that users are also allowed to switch to manual mode and adjust LED brightness and blinds opening-closing themselves if they are not satisfied with automatic dimming.

Initial UV($\mu\text{W}/\text{cm}^2$)	Blinds status	Initial illuminance(lux)	Final illuminance(lux)
8	open	200	476
50	partly closed	700	545
85	fully closed	900	562

Table 1 The Results of the Validation Experiment

Feasibility and workability are validated through prototype testing in a simulated library environment. An adjustable light source (including UV-A) simulated natural sunlight variations. Target illuminance (500 lux) was set. Light/UV intensity was gradually increased to trigger automatic shading and LED dimming. The results are shown in Table 1. It can be concluded that integrated UV sensor and photoresistor works. System auto-regulates for lighting. And therefore core functionality and energy savings are validated.

VI. Objectives

- Connecting circuits: Enables sensor data to be transmitted, and the brightness of LED bulbs and the opening and closing degree of blinds can be controlled.

Connecting circuits to enable sensor data transmission and control of LED bulb brightness and blind opening or closing degree ensures that the system's core components work together, making accurate device regulation based on real-time sensor inputs possible.

- Programming Arduino AVR: Enable it to receive data from different sensors and then control the output (LED bulbs and blinds)

Programming the Arduino AVR to receive data from different sensors and control outputs like LED bulbs and blinds enables the system to respond automatically to environmental changes, laying the groundwork for its smart adjustment function.

- Building Simulation Environment: Build a shed with blinds to simulate a library environment.

Building a shed with blinds to simulate a library environment aims to create a realistic test setting, which allows the system to be tested under library-like light conditions without using an actual library space.

- Optimize the entire system

The computational analysis provides a means of analyzing multiple solutions cost effectively because all significant parameters may be analyzed to estimate the resulting effective system response.

VII. Tasks

To ensure the smooth implementation of the Lighbrary project, we have designed a sequential task workflow.

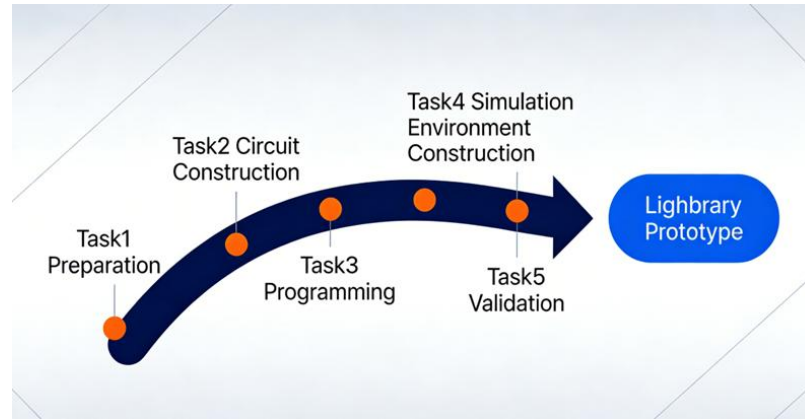


Figure 5 Task Flow Diagram

This diagram illustrates the project's progressive workflow: starting from preliminary preparation, it sequentially covers circuit construction, programming, simulation environment building, and overall debugging, to complete the development of the Lighbrary prototype.

- Task 1: Preparation

During this task, we do an overall structure design and prepare necessary resources to lay a solid basis for subsequent implementation.

We confirm core objectives of the Lighbrary system (adaptive LED dimming, blind control, touchscreen interaction), compile a standardized Bill of Materials (BOM) covering hardware (Arduino Uno, photoresistor, UV sensor, LED bulbs, stepper motor) and auxiliary materials (breadboards, jumper wires) and finalize team task allocation and formulate a preliminary 6-week schedule.

After the completion of Task 1:

- Verified and categorized project BOM
- Team task allocation table & preliminary schedule.
- Task 2: Circuit Construction

During this task, we build physical connections for the hardware modules of the Lighbrary system, establishing the data transmission and control links between sensors, controllers, and actuators, as shown in Figure 6.

After the completion of Task 2:

- Wired & functional circuit system

- Task 3: Programming

During this task, we develop control logic code for the Lighbrary system to realize its adaptive adjustment functions.

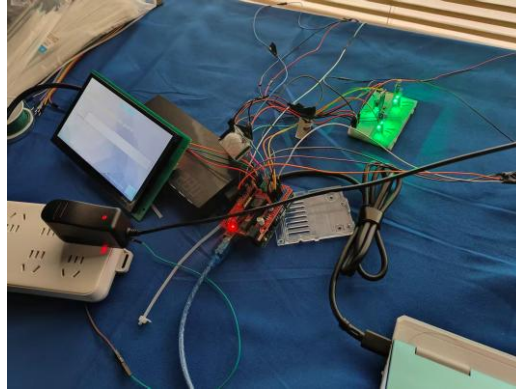


Figure 6: a part of the circuit

We write Arduino code to collect real-time data (illuminance from photoresistor, radiation intensity from UV sensor), develop control algorithms (xxxxxx), and debug code to xxxxx.

After the completion of Task 3:

- Functional Arduino control code (including data collection & device control logic)
- Task 4: Simulation Environment Construction

During this task, we build a simulated library environment to replicate real scenarios and test the adaptive performance of the Lighbrary system.

We fabricate a 1.5m×1.8m shed (iron frame + light-blocking curtain) as the simulated space, install components in the shed (LED bulbs, photoresistor & UV sensor, capacitive touchscreen on the outer surface), and integrate the assembled circuit into the shed (fixed with cable ties and tapes to avoid loose connections).

After the completion of Task 4:

- Complete simulated library environment (shed with installed components)
- Task 5: Validation

During this task, we conduct comprehensive testing and optimization to ensure the Lighbrary system meets design requirements for stability, functionality, and user experience.

We perform scenario tests (verifying blind closing & LED dimming in glare scenarios, LED brightness enhancement in low-light scenarios), optimize system performance and fix identified issues (e.g., LED brightness fluctuations) via code and mechanical calibration.

After the completion of Task 4:

- Optimized full-system code (with improved stability & real-time performance)

VIII. Discussion

This library-adaptive smart lighting and shading system is with clear strengths and limitations. Its core merits lie in the low-cost simulated test environment, seamless Arduino-driven automatic control of lights and blinds, and optimized stable operation, all of which effectively achieve glare reduction and energy-saving goals.

Yet the design is far from perfect: the small-scale simulation cannot reflect the complexity of real libraries, and the lack of personalized settings fails to meet diverse user needs. Minor sensor response delays also affect real-time performance during testing. With more development time, two critical future improvements will be prioritized.

First, adding personalized brightness settings would let individual users or specific zones set preferred light levels via a simple interface; the system would then remember these preferences and adjust automatically, instead of applying uniform lighting across the entire space. This customization would address the varying needs of library users, such as those who need brighter light for reading and those who prefer dimmer conditions for focused work.

Second, optimizing performance and adapting the system to actual libraries involves refining sensor to eliminate response delays and expanding the control algorithm to support multiple independent zones. Performance optimization would include upgrading the Arduino program to process sensor data faster and adding redundant sensors to ensure reliability, while adaptation to large libraries would require dividing the space into interconnected sub-systems, each managing its own blinds and lights based on local light conditions.

To showcase the system's value to sponsors, a live demo could be held to display its glare-blocking and energy-saving effects in the simulated setup, proving its practical value for real library applications.

IX. Conclusion

The Lighbrary project developed an IoT-based adaptive lighting system to enhance reading comfort and promote energy efficiency in libraries. It addresses the common issues of inconsistent illumination and energy waste by automatically adjusting light output based on real-time sensor data. Light sensors maintain optimal, consistent brightness at reading surfaces, while occupancy sensing ensures lights are dimmed in unoccupied areas. This automation is complemented by user-friendly touchscreen controls, allowing for personal preference overrides.

The innovation lies in the holistic integration of these features into a single, responsive system that prioritizes both human comfort and sustainability. The significance is substantial:

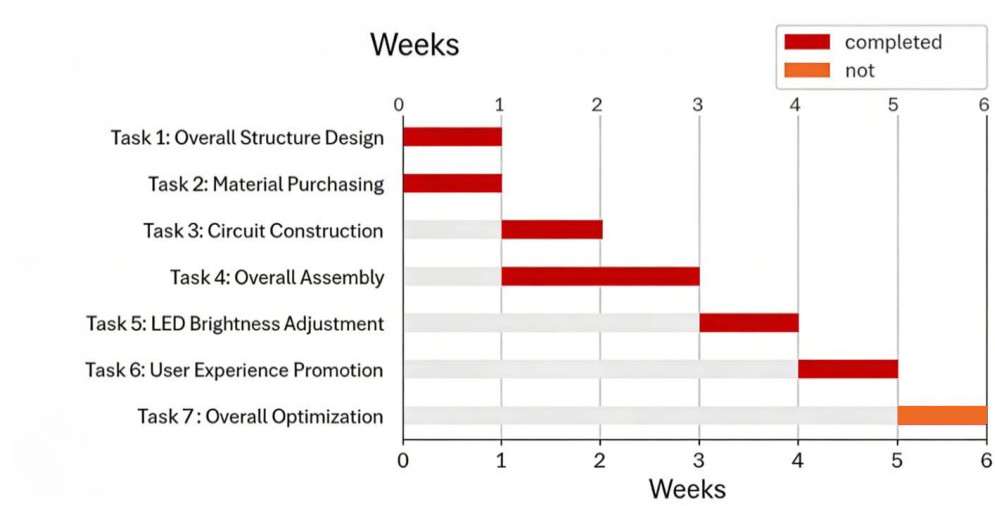
Libraries benefit from significantly reduced operational costs. The adaptive dimming and occupancy-triggered power management cut down unnecessary electricity consumption, translating to lower monthly utility bills and reduced long-term maintenance expenses for lighting fixtures due to less frequent use and wear. This cost-saving effect enables libraries to reallocate financial resources to core services such as book acquisitions, digital resource upgrades, and reader education programs.

Readers can enjoy a superior, tailored lighting environment for learning and focus. The consistent, sensor-calibrated brightness eliminates glare on reading materials and eye strain caused by over-bright or dim lighting, creating a more ergonomic and pleasant space for prolonged reading, studying, and research. The customizable touchscreen controls further empower users to adjust lighting to their individual visual preferences, whether for intensive academic work or casual reading, thereby boosting overall reading efficiency and user satisfaction.

Environmentally, the system directly reduces carbon footprint, supporting broader institutional and societal sustainability goals. By curbing electricity use—especially during off-peak hours or in underutilized library zones—the system decreases the reliance on fossil fuel-powered electricity generation, cutting down greenhouse gas emissions associated with power production. This aligns libraries with global carbon neutrality initiatives and sets an example of green infrastructure adoption for public and educational institutions, inspiring other facilities to implement similar energy-saving solutions.

The project demonstrated that intelligent infrastructure can simultaneously achieve significant economic, environmental, and social impact. Economically, it drives cost efficiency; environmentally, it advances sustainability; and socially, it enhances the accessibility and quality of public cultural spaces, fostering a more inclusive and reader-friendly community hub. This triple-win outcome proves that smart technology integration is not just a modernization trend but a practical approach to balancing operational needs, environmental responsibility, and public welfare.

X. Schedule



XI. Bill of Materials

Item	Quantity	Cost (RMB)	Hyperlink
Venetian Blinds PVC (Customized)	1	135.99	https://e.tb.cn/h.Szoboje7dwnVdFe?tk=g06MfKR4BQ1
Ultraviolet Detection Module	1	19.85	https://detail.tmall.com/item.htm?id=620787278032
Ultraviolet Lamp	1	49.3	https://detail.tmall.com/item.htm?id=921123843338
Capacitive Touch Screen	1	243	https://item.taobao.com/item.htm?id=663412298335
Electron Beam	5	23.25	https://detail.tmall.com/item.htm?id=746062545541
Jumper Wire	1	16.8	https://detail.tmall.com/item.htm?id=669570807944
Stepper Motor Driver Board TB6600	1	39.16	https://detail.tmall.com/item.htm?id=555488391549
Original Arduino Uno	1	106	https://detail.tmall.com/item.htm?id=775283572218
Variable Brightness LED Light	5	54	https://item.taobao.com/item.htm?id=676013432739
Breadboard	8	14.64	https://e.tb.cn/h.71H4bS7Hui5O3Py?tk=smSfftPu3ax
Lampshade	1	9.8	https://item.taobao.com/item.htm?id=654957085699
Thickened Light-blocking Bed Curtain (1.5m×1.8m)	1	178	https://e.tb.cn/h.7Y3g3sS5enUk4se?tk=bnJtfli
Nylon Cable Tie	1	22.95	https://e.tb.cn/h.7Y3tbkhEprsUK20?tk=wMqFfIRUDa2
12V Lithium Battery	1	42	https://e.tb.cn/h.7c3va9N8HEZpAPk?tk=VQ0oftPt0Yr
Arduino Power Expansion Board	1	60	https://e.tb.cn/h.71HfLnaXbBvUNdN?tk=4mEUftPulDE
Total	/	1014.74	/



Key Personnel

Congyu Yang is responsible for program design and other major technical aspects.

Chen Liang, the team leader of the group, mainly distributed missions to each team member and participated in part of the coding part, the connection of circuits, preparations of presentations and the final setup.

Kuangdi Li participated in the procurement of materials, the preparation of presentations, the drafting of reports, as well as the setup of simulation environment and the assembly of the entire system.

Shuhan Guo participated in the procurement of materials, the preparation of presentations, the drafting of reports, as well as the setup of simulation environment and the assembly of the entire system.

XII. References

- [1] R. Kurniawati, M. Mardji, and A. Kurniawan, "Effect of Light Intensity On Eye Fatigue," in Advances in Health Science Research (AHSR), vol. 7, Proc. 2nd Int. Conf. on Sports Sciences and Health 2018 (ICSSH 2018), 2018.
- [2] [Zhongneng Media Energy Security New Strategy Research Institute], "China Energy Big Data Report (2025) - Power Section," bjx.com.cn (Polaris Power News), 4 Jul. 2025. [Online]. Available: <https://news.bjx.com.cn/html/20250704/1449662.shtml>. [Accessed: 02 Dec. 2025].

Appendix

```

25 if (n_TempChar == 64) { m_ReciverVarInt = mySerial.parseInt();}
26 else if (n_TempChar == 35) { m_ReciverVarFloat = mySerial.parseFloat();}
27 else if (n_TempChar == 36) { mySerial.read();}
28 m_ReciverVarString = mySerial.readStringUntil(13);}else return 0;
29 return 1;}
30 return 8;}
31 void procedure() {
32   if (analogRead(A4) < 100 && analogRead(A4) >= 0) {
33     delta = 0 - switch2;
34     switch2 = 0;
35   }
36   } else if (analogRead(A4) < 200 && analogRead(A4) >= 100) {
37     delta = 1 - switch2;
38     switch2 = 1;
39   } else if (analogRead(A4) < 400 && analogRead(A4) >= 200) {
40     delta = 2 - switch2;
41     switch2 = 2;
42   } else if (analogRead(A4) < 600 && analogRead(A4) >= 400) {
43     delta = 3 - switch2;
44     switch2 = 3;
45   } else if (analogRead(A4) < 800 && analogRead(A4) >= 600) {
46     delta = 4 - switch2;
47     switch2 = 4;
48   } else if (analogRead(A4) <= 1024 && analogRead(A4) >= 800) {
49     delta = 5 - switch2;
50     switch2 = 5;
51   }
52   if (delta >= 0) {
53     direction = 1;
54   }
55   } else {
56     direction = 0;
57     delta = 0 - delta;
58   }
59   }
60   turning();
61   }
62   }
63   void turning() {
64     if (direction == 1) {
65       digitalWrite(10,HIGH);
66     }
67     } else {
68       digitalWrite(10,LOW);
69     }
70   }

```

```

71   for (int i = (1); i <= (delta * 100); i = i + (1)) {
72       digitalWrite(7,HIGH);
73       delayMicroseconds(1500);
74       digitalWrite(7,LOW);
75   }
76   }
77   }
78   }
79   void setup(){
80       light = 100;
81       detection = 0;
82       direction = 1;
83       switch2 = 0;
84       delta = 0;
85       Mode = 0;
86       Brightness = 0;
87       Serial.begin(9600);
88       mySerial.begin(9600);
89       pinMode(8, INPUT);
90       pinMode(3, OUTPUT);
91       pinMode(10, OUTPUT);
92       pinMode(7, OUTPUT);
93       digitalWrite(10,HIGH);
94       digitalWrite(7,LOW);
95       pinMode(A1, INPUT);
96       pinMode(A2, INPUT);
97       pinMode(A3, INPUT);
98       pinMode(A4, INPUT);
99       pinMode(3, OUTPUT);
100  }
101
102  void loop(){
103      if (ATfMessageService(2)) {
104          if (m_ReceiverBodyID == 100) {
105              Mode = 1;
106              mySerial.print("@GUIS ");
107              mySerial.println(Mode);
108          }
109          else if (m_ReceiverBodyID == 101) {
110              Mode = 0;
111              mySerial.print("@GUIS ");
112              mySerial.println(Mode);
113          }
114          else if (m_ReceiverBodyID == 102) {
115              Brightness = m_ReceiverVarInt;
116          }
117          else if (m_ReceiverBodyID == 103) {
118          }
119          else if (m_ReceiverBodyID == 104) {
120          }
121      }
122      detection = (analogRead(A1) + (analogRead(A2) + analogRead(A3))) / 3;
123      Serial.println(detection);
124      if (digitalRead(8) == 1) {
125          if (Mode == 0) {
126              if (detection <= 400 + Brightness) {
127                  if (light - 7 < 30) {
128                      light = 0;
129                  }
130                  else {
131                      light = light - 7;
132                  }
133              }
134              else if (detection >= 450 + Brightness) {
135                  light = min(light + 7, 255);
136                  analogWrite(3, light);
137              }
138          }
139          else {
140          }
141      }
142      else {
143          analogWrite(3, 0);
144      }
145      procedure();
146      delay(100);
147  }
148  }
149  }

```

Figure 7: code